MILITARY SPECIFICATION

HAT, CAMOUFLAGE PATTERN: DESERT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a hat made of cotton/nylon twill material dyed in a six-color camouflage desert pattern.

1.2 Classification. The hat shall be of one type in the following sizes as specified (see 6.2).

Schedule of sizes

6 3/8	6 7/8	7 3/8
6 1/2	7	7 1/2
6 5/8	7 1/8	7 5/8
6 3/4	7 1/4	7 3/4
		7 7/8

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 8415

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SPECIFICATIONS

FEDERAL

V-L-61	-	Laces, Nylon
KK-L-271	-	Leather, Cattlehide, Strap, Vegetable Tanned
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered
MIL-C-43836	-	Cloth, Interlining, Non-woven
MIL-C-44034	-	Cloth, Twill, Camouflage Pattern, Cotton and Nylon for Desert Uniform

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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THE COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Reference of America

(Copies should be obtained from the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 Materials (see 6.4).

3.2.1 Basic material. The basic material shall be a blended cotton/nylon, twill cloth dyed in the US Army six-color camouflage desert pattern conforming to MIL-C-44034.

* 3.2.1.1 Basic material seconds. The cloth for side crown shall be basic material or printed seconds cloth. Printed seconds cloth shall be a cloth rejected only for defects pertaining to color, infrared reflectance, or camouflage print pattern of MIL-C-44034. If printed seconds are used, they must be marked on the face side with stripes, 3 inches apart, the full width, running in a warpwise direction for the length of the entire roll. The stripes shall be 1/8 to 1 inch wide and shall be applied by the cloth finisher. The color of the stripes shall approximate any one of the camouflage print colors, and colorfastness to crocking shall be no lower than 6.5 (Munsell Scales).

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3.2.2 Brim binding. The brim binding shall be bias-cut from the basic material (see 3.2.1).

3.2.3 Brim interlining and headband stay piece. The brim interlining and headband stay piece shall be a white, nonwoven, continuous filament polyester cloth conforming to type I, class 2 of MIL-C-43836.

3.2.4 Laces, nylon. The nylon lace for the chin strap shall be Tan 337 conforming to type II, class 2 of V-L-61 except that the requirements for length and tipping shall not apply.

3.2.5 Keeper. The chin strap keeper shall be made from vegetable-tanned cattlehide strap leather, 6/64 to 8/64 inch thick, natural in color, conforming to type I, II, III, or IV, class 1 of KK-L-271 except that the requirements for the treatment for fungicide paranitrophenol, crocking, cracking, breaking force, elongation, chloroform extract, water-soluble material, total ash, and degree of tannage shall not apply. A plastic fixlock keeper may be substituted for the leather keeper.

3.2.6 Thread. The thread for seaming and stitching the hat shall be Khaki, P-1, C.A. 66019 polyester core cotton covered and rayon covered, ticket Nos. 50, 3 ply, and 70, 2 ply conforming to MIL-T-43548.

3.2.6.1 Colorfastness. The dyed thread shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to laundering.

3.2.7 Labels. Each hat shall have a combination identification-size label and an instruction label conforming to DDD-L-20. The label color shall be Khaki, Cable No. 70188 of the Color Association of the United States, Standard Color Reference of America and shall show fastness to laundering in accordance with DDD-L-20 requirements.

3.2.7.1 Identification-size label. The identification-size label shall conform to type VI, class 4.

3.2.7.2 Instruction label. The instruction label shall conform to type VI, class 3 and include the following information.

CAUTION

Initial loose fit required
for continued comfort.

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LAUNDERING

Spot wash badly soiled areas.
Wash in hot water (140°F)
using soap or detergent.
Machine spin or squeeze by hand.
Machine dry at moderate heat
or hang to dry.

DO NOT STARCH

Suspend insect net from top of brim, drape
headnet over shoulders and attach front
loops to breast pocket buttons.

DO NOT REMOVE THIS LABEL

3.2.8 Ventilators, with washers. The screen type ventilators with washers shall be brass, have a black chemical finish, a 9/16 inch outside rim diameter, and a 7/16 inch outside diameter screening, conforming to Stimpson Company, No. 53 or equal (see 6.5). The contractor shall submit a certificate of compliance for approval (see 4.2.1.1).

* 3.2.9 Adhesive (when applicable). The adhesive for holding the top and bottom brim and brim interlining prior to performing the quilting stitching of the brim shall be a synthetic resin emulsion type adhesive with good machining properties. The adhesive shall have a viscosity of 2000 to 2500 C.P.S. when evaluated on a Brookfield RVF viscosimeter (see 6.6) at 77°F using a number 3 spindle at 20 rpms, a solids content of 57 ± 2 percent, a pH of 4.0 to 6.0 and shall weigh 9.2 ± 0.2 pounds per gallon. The adhesive shall be white when wet and colorless when dry. When option to use adhesive is selected the contractor shall submit a certificate of compliance for approval (see 4.2.1.1).

3.3 Design. The hat is a lightweight soft crowned camouflage model with a standard width quilt stitched brim and chin strap. The hat is lined with self material.

3.4 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government. The working patterns shall be identical to the standard patterns which shall not be altered in any way. The patterns provide 3/16 inch for seaming side lining to crown, 3/8 inch for seaming brim to sides, and 1/4 inch for seaming all others.

* 3.4.1 List of pattern parts. The components of the hat shall be cut from the materials specified and according to the number of parts listed in table I.

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TABLE I. List of pattern parts

Material	Pattern nomenclature	Cut parts
Basic material	Top crown	1
	Top side	1
	Bottom side	1
	Top and bottom side crown (alternate construction)	1
	Side lining	1
	Top brim 1/	1
	Bottom brim 1/	1
Non-woven material	Brim interlining	1
	Headband stay	1

1/ The top brim and bottom brim are the same pattern part.

3.5 Construction. The construction shall conform in all respects to the requirements specified in table II and herein. Figure 1 is furnished solely for guidance and information. Should variation from the written document appear in figure 1, the document shall govern.

3.5.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the hat.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

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3.5.1.1.1 Repairs of type 301 stitching.

a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 101 and 401 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. All repairs shall be in accordance with 3.5.1.1.1a. and 3.5.1.1.1b. Repairs to stitch type 401 may be accomplished by use of stitch type 301.

3.5.2 Bartacking. Bartacking shall be 1/4 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 8 to 14 stitches. Bartacks shall be free from thread breaks and loose stitching.

3.5.3 Automatic stitching. Automatic machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping, or backstitches are used to secure the ends of the stitching.

3.5.4 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the hats.

* 3.6 Manufacturing operations requirements. The hat shall be made in accordance with the operation requirements specified in table II. The contractor is not required to follow the exact sequence of operations listed, unless otherwise specified. Seam allowances required to be turned and seam turn-up of side and lining may be precreased. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished hat.

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TABLE II. Manufacturing Operations Requirements

No.	Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread	
					Needle	Bobbin/ looper
1.	Cutting.					
	a. Cut all parts in strict accordance with patterns furnished, which show size, directional lines (warp or filling direction) and notches for proper assembly of parts.					
	b. Cut all parts of basic material from one piece of material except side lining and brim binding which may be cut from ends. The side lining, when cut from ends, shall approximate the shade of the respective basic material.					
	c. Cut the brim binding on the bias, not less than 1-1/8 inches wide. The brim binding may be placed on rolls.					
	d. Cut the lace for the chin strap with a heated blade to fuse ends of straps. The cut lengths shall finish 25 ± 1 inch.					
2.	Replacement of damaged parts.					
	Care shall be exercised during spreading, cutting and manufacturing operations to assure that material defects and damages, as specified in 4.2.2, are excluded and replaced with non-defective and properly matched material.					

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TABLE II. Manufacturing Operations Requirements - Continued

No.	Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/looper
3.	Marking. Mark, ticket, or bundle all cut parts to ensure a uniform shade, and size throughout the hat. Any method of marking may be used except: a. Corrosive metal fastening devices. b. Sew-on tickets. c. Adhesive type tickets which leave an impression or traces of paper or adhesives on the material after removal of the ticket.					
4.	Make chin strap keeper and chin strap. a. Cut leather piece for chin strap keeper, 3/4 ±1/8 inch long by 1/2 ± 1/8 inch wide. b. Punch two holes, equally spaced, in keeper to accommodate chin strap. The holes shall be of sufficient length and width to ensure proper function of the keeper, i.e., keeper shall not move without exertion of a slight pressure. c. Thread chin strap through holes in keeper.					

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TABLE II. Manufacturing Operations Requirements - Continued

No.	Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/looper
5.	Make brim.					
	Finished appearance. The brim shall be made of two plies of basic material interlined with one ply of non-woven material and shall finish without twists or puckers. The brim shall be quilted with parallel lines of stitching and shall finish 2-3/8 ± 1/8 inches (see table III) and finish with an upward curl. The contractor has the option to use adhesive to join the top and bottom brim and brim interlining prior to performing the quilting stitch operation of the brim.					
	a. (1) Superimpose outside edge of bottom brim and brim interlining with notches matching, and stitch together 1/16 to inch from outside edges.	101 or 301 or 401	SSa-1	6-14	50 50 50	 50 70
	(2) Superimpose outside edges of top brim and brim interlining with notches matching and stitch 1/16 to 1/8 inch from outside edges through all plies.	101 or 301 or 401	SSa-1	6-14	50 50 50	 50 70
	or					
	b. Superimpose outside edges or bottom brim, brim interlining, and top brim with notches matching and stitch together 1/16 to 1/8 inch from outside edges in one operation.	101 or 301 or 401	SSa-1	6-14	50 50 50	 50 70

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TABLE II. Manufacturing Operations Requirements - Continued

No. Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread	
				Needle	Bobbin/ looper
5. Make brim. - continued					
c. Stitch bottom brim to top brim along inner edge adjacent to brim interlining.	301		8-10	50	50
d. Single or double stitch brim plies together with the row of stitches starting from outside edge of brim and continue stitching for the entire brim width. The rows of stitching shall be spaced 3/16 inch to 1/4 inch apart. The needle thread shall finish on top brim in the finished hat.	301		8-10	50	50
e. Turn under one end of binding 1/4 inch and superimpose edges of binding on underside of brim and stitch binding to outside edge of brim 3/16 to 1/4 inch from edge, holding binding tight. The binding shall start and finish at center back (± 1/2 inch tolerance) with the turned end of binding lapped not less than 1/2 inch on the opposite end of binding.	301 or 401	BSg-2(a)	10-14	50 50	50 70
f. Turn binding over edge of brim, turn in raw edge and stitch 1/16 to 3/32 inch from edge through binding and brim. Binding shall finish not more than 3/8 inch wide on top side of brim, and shall be so attached that brim will curve upward 3/4 to 1 inch.	301	BSg-2(b)	10-14	50	50
or					
g. As an alternate the binding may be attached to the brim in one operation, using a folder.					

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TABLE II. Manufacturing Operations Requirements - Continued

No.	Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/looper
6.	Join sides.					
	a. Join top side to bottom side with front notches matching.	301 or 401	LSq-2(a)	10-14	50 50	50 70
	b. Turn down bottom side and raise stitch bottom side, 1/16 to 3/32 inch from outside edge.	301	LSq-2(b)	10-14	50	50
	or					
	Finished appearance. The finished mock seam shall be uniform in width, without distortion, and be accomplished only by the use of a folder.					
	c. Position side piece in folder, pass piece through folder in accordance with marks on pattern. The outside folded edge of the mock seam shall face the top of the crown with stitching 1/16 to 3/32 inch from folded edge.	301 or 401	OS E-1	10-14	50 50	50 70
	d. Join center back seam of top and bottom sides. Open seam allowance and stitch through the turned edges 1/16 to 3/32 inch from each side of joining seam line.	301 or 401 and 301	SSa-1 SSz-3(b) shows finished seam	10-14 10-14	50 50	50 50

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TABLE II. Manufacturing Operations Requirements - Continued

No.	Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
	Join sides. - continued					
	e. Stitch the instruction label on the inside of hat to the side lining on left side of hat, positioned 3/4 to 1 inch from back raw edge. Stitch label on all four sides.	301	LSbj-1	10-14	50	50
	f. Join center back seam of side lining and open seam allowances.	301 or 401	SSa-1	10-14	50 50	50 70
7.	Join sides to crown.					
	Finished appearance. The top side piece shall finish on top of crown and shall be free of twists, puckers, pleats, or raw edges.					
	a. Join side lining to inside of crown, matching front notches and placing back seam in line with back notch on crown. Stitching shall not show on outside.	101 or 301 or 401	SSa-1	10-14	50 50 50	 50 70
	b. Join top side to outside of crown, matching front notches and placing back seam in line with back notches on crown.	301 or 401	LSbp-2(a)	10-14	50 50	50 70
	c. Turn down top side and raise stitch top side, 1/16 to 3/32 inch from edge.	301	LSbp-2(b)	10-14	50	50
	NOTE: The stitching specified in operations b and c shall not show on inside of hat.					

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TABLE II. Manufacturing Operations Requirements - Continued

No.	Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
8.	Join sides to brim.					
	a. Position ends of the chin strap to underside of brim at each side notch and tack in position.	301 or 401 or bartack		6-14	50 50	50 70
	b. Position bottom side piece to top of brim matching front notches and placing back seam in line with back notch of brim and raw edges even and stitch side piece to brim, not through brim interlining.	301 or 401	LSq-2(a)	10-14	50 50	50 70
	c. Turn up bottom side piece and edge stitch side piece 1/16 to 3/32 inch from edge.	301 or 401	LSq-2(b)	10-14	50 50	50 70
	d. Position headband stay piece to inside of side lining with bottom edge of stay piece 1/2 inch from bottom edge of lining and even with back seam of lining. Stitch top edge of stay piece to lining 1/16 to 1/8 inch from edge of stay piece.	301 or 401	LSq-2(b)	10-14	50 50	40 70
	e. Turn up bottom raw edge of lining over bottom edge of stay piece.					
	f. Position side lining to bottom of brim matching front notches and back seam in line with back notch of brim and stitch side lining to brim, 1/16 to 3/32 inch from turned edge, catching bottom edge of size label in the stitching. The label shall extend upward into the crown and be positioned to right side of center back seam.	301		10-14	50	50

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TABLE II. Manufacturing Operations Requirements - Continued

No.	Requirements	Stitch type	Seam and stitching type	Stitches per inch	Thread: Needle	Thread: Bobbin/ looper
8.	Join sides to brim. - continued NOTE: The lining shall not be full or twisted on the finished hat nor shall it twist or cause fullness to the outside of the finished hat. The lining shall not extend below the brim.					
9.	Blocking. Block crown on proper sized heated blocks and flange brims with metal dies on hydraulic press with heat and pressure to properly set brims.					
10.	Attach ventilators. Pre-punch holes through side and lining, as indicated by marks on pattern, and attach a ventilator at each hole with washer on inside of hat.					
11.	Clean hats. a. Trim all thread ends. b. Remove all loose threads, spots, stains, shade markings, and size markings.					

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3.7 Measurements. The finished hat shall conform to measurements for each respective size, as shown in table III.

TABLE III. Finished measurements (in inches)

Size	Inside hat (head measurements) 1/ ±1/4 inch	Crown height at center front 2/ ±1/4 inch	Crown height at center back 3/ ±1/4 inch	Brim width 4/ ±1/8 inch
6 3/8	20-1/2	2-3/4	3-1/2	2-3/8
6 1/2	20-7/8	2-3/4	3-1/2	2-3/8
6 5/8	21-1/4	2-3/4	3-1/2	2-3/8
6 3/4	21-5/8	2-3/4	3-1/2	2-3/8
6 7/8	22	2-3/4	3-1/2	2-3/8
7	22-3/8	2-3/4	3-1/2	2-3/8
7 1/8	22-3/4	2-3/4	3-1/2	2-3/8
7 1/4	23-1/8	2-3/4	3-1/2	2-3/8
7 3/8	23-1/2	2-3/4	3-1/2	2-3/8
7 1/2	23-7/8	2-3/4	3-1/2	2-3/8
7 5/8	24-1/4	2-3/4	3-1/2	2-3/8
7 3/4	24-5/8	2-3/4	3-1/2	2-3/8
7 7/8	25	2-3/4	3-1/2	2-3/8

1/ Measure along bottom edge of side lining.

2/ From base of side to crown seam on outside of center front.

3/ From base of side to crown seam on outside of center back.

4/ From base of side to outer bound edge of brim.

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3.8 Workmanship. The finished hat shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

* 4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.2.1.1 Component and material certification. Unless otherwise specified, certificates of compliance will be acceptable as evidence that the ventilators in 3.2.8 and the adhesive in 3.2.9 conform to the specified requirements.

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* 4.2.2 End item visual examination. The end items shall be examined for the defects listed in table IV. The lot size shall be expressed in units of hats. The sample unit shall be one hat. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 10.0 for major and minor A combined defects, and 15.0 for total (major, minor A, and minor B combined) defects.

TABLE IV. End item visual defects

Examine	Defect	Classification		
		Major	Minor A	Minor B
Blocking	Omitted	X		
	Poorly blocked, causing distortion on crown or brim; hat not symmetrical, affecting appearance		X	
Cleanness	Any spot or stain on outside			X
	Thread ends not trimmed, loose threads, or one or more shade or size marking not removed			X
Component and assembly	Any component part or required operation omitted (unless otherwise classified herein)	X		
	Any operation not performed as specified (unless otherwise classified herein)			X
Cutting	Any component part not cut in strict accordance with directional lines indicated on pattern or not cut in accordance with document requirements	X		
Ventilators	One or more omitted or added	X		
	Broken or bent	X		
	One or more improperly clinched, i.e., loose or raw edges exposed around edges of ventilator			
	- on outside	X		
	- on inside		X	

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TABLE IV. End item visual defects - Continued

Examine	Defect	Classification		
		Major	Minor A	Minor B
Ventilators - continued	Side puckered, pleated or full in clinching one or more ventilators		X	
	One or more ventilators with washer on outside of hat	X		
Label (identification-size or instruction)	Missing, incorrect, or illegible	X		
	Not securely attached		X	
Material defects and workmanship damages	Any hole or shade bar	X		
	Any defect such as a smash, multiple float, or slub	X		
	Cut, tear, rip, mend, or burn	X		
	Needle chew	X		
Seams and stitching, general	Seam or component part twisted, puckered, or pleated		X	
	Part of material caught in any unrelated operation or stitching	X		
	Ends of stitching not secured as specified		X	
	Thread break (all stitch types) stitched less than 1/2 inch beyond each end of break			X
Gage of stitching and seam allowances	Irregular, i.e., unevenly gaged, or corresponding stitchings not uniformly gaged (to be scored only when condition exists along major portion of seam)		X	
	Width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified			X

NOTE: The above defects to be scored only when the condition exists on major portion (i.e., over one half or more) of seam.

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TABLE IV. End item visual defects - *Continued*

Examine	Defect	Classification		
		Major	Minor A	Minor B
Open seams	Open seams (one or more broken or two or more consecutive skipped or runoff stitches constitute an open seam):			
	Any seam open on outside:			
	- up to 1/4 inch inclusive			X
	- more than 1/4 inch	X		
	Any seam open on inside of hat:			
	- up to 1/4 inch inclusive			X
	- more than 1/4 inch to 1 inch inclusive		X	
	- more than 1 inch	X		
Raw edges	Any raw edge on outside of hat:			
	- 1/4 to 1/2 inch inclusive		X	
	- more than 1/2 inch	X		
	Any raw edge on inside of hat more than 1/4 inch			X
	NOTE: Raw edges not securely caught in stitching shall be classified as an open seam.			
Runoffs	On joining seams, when resulting in an open seam, score as "open seam"			
	On binding stitching:			
	- 1/4 to 1/2 inch inclusive			X
	- over 1/2 inch		X	

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TABLE IV. End item visual defects - Continued

Examine	Defect	Classification		
		Major	Minor A	Minor B
Seam and stitch types	Not specified seam or stitch type		X	
	Looper thread on outside (when 401 stitch is used)	X		
	Chain of 101 stitch type exposed on outside of hat	X		
Stitch tension	Loose tension resulting in a loose seam		X	
	Loose tension on top quilting stitches or edge stitching, resulting in loosely exposed lower thread on top thread		X	
	Tight tension (stitches break when strain is applied to the direction of seam)	X		
Stitches per inch (to be scored only when the condition exists on major portion of seam or stitching)	Less than minimum specified		X	
	More than maximum specified			X
Shaded parts	Badly shaded parts (except those cut from ends)	X		
	Parts permitted to be cut from ends badly shaded			X
Brim	Twisted, stretched, or poorly shaped	X		
	Rows of quilting stitches less than 3/16 inch or more than 1/4 inch apart		X	

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TABLE IV. End item visual defects - Continued

Examine	Defect	Classification		
		Major	Minor A	Minor B
Brim - continued	Needle thread of quilting stitching finished on underside of brim			X
	Fullness between rows of quilting stitches		X	
	Binding, twisted, puckered, or pleated	X		
	Outside end of binding not turned under or not within 1/2 inch of center back of brim			X
	Binding finished more than 3/8 inch wide on top of brim for a distance of 2 inches or more		X	
	Irregular in width by more than 1/2 inch inclusive		X	
	Ends of binding lapped less than 1/2 inch		X	
	Brim curving upward less than 3/4 inch or more than 1 inch		X	
Crown and sides	Crown twisted or distorted		X	
	Side twisted or distorted	X		
	Side lining twisted or distorted			X
	Lining extending below edge of brim			X
	Side back seam and side lining back seam, not seam on seam, by more than 1/8 inch		X	
	Stitches joining top side to crown exposed on inside of hat		X	
	Stitches joining side lining to crown exposed on outside of hat		X	
	Joining seams of top side to bottom side out of alignment by more than 1/16 inch at back seam		X	

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TABLE IV. End item visual defects - Continued

Examine	Defect	Classification		
		Major	Minor	
			A	B
Chin strap and keeper	Chin strap missing	X		
	End of chin strap not fused		X	
	Keeper missing	X		
	Keeper less than 5/8 inch or more than 7/8 inch long; or less than 3/8 inch or more than 5/8 inch wide			X
	Holes in keeper too small or too large not permitting chin strap to function properly, i.e., falls freely from set position or keeper immovable	X		
	Chin strap measuring less than 24 inches or more than 26 inches long			X
	Chin strap twisted			X

4.2.3 End item dimensional examination. The end items shall be examined for the dimensions listed in table III. Any dimension not within the specified tolerance shall be scored as a defect. The lot size shall be expressed in units of hats. The sample unit shall be one hat. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.2.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Material	Any component missing, damaged, or not as specified

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Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, inadequate stapling, or improper taping Bulged or distorted container Open and noncontinuous heat-sealed seams and closure of polyethylene bag Incorrectly fabricated bag Alternative closure of polyethylene bag not provided with a twist tie
Content	Number of intermediate packs per shipping container is more or less than required Number of hats per intermediate pack is more or less than required 1/

1/ For this defect, one intermediate pack from each container in the sample shall be examined.

* 4.2.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

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5.1.1 Level A. Each hat shall be firmly rolled along the lengthwise dimension of the hat. Each hat shall be inserted into a snug-fitting polyethylene film bag of 0.00125 inch thickness (± 25 percent tolerance). The bag may be fabricated from polyethylene film tubing or sheeting. The bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and to the formed edges of the bag. The final closure of the bag shall be heat sealed with the seal made as close as possible to the open end. Alternatively, bag closure may be accomplished by a twist tie method. Prior to or during the closure operation, excess air within the bag shall be expelled.

5.1.1.1 Intermediate pack. Twelve hats of one size only, unit packed as specified in 5.1.1, shall be placed in an intermediate fiberboard box conforming to style FOL (end-opening), grade W6c or W6s of PPP-B-636. The hats shall be placed flat, one in length, one in width, and twelve in depth. Inside dimensions of the fiberboard box shall be 14-1/2 inches in length, 4 inches in width, and 13 inches in depth. Each fiberboard box shall be closed in accordance with method IV as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4.

* 5.1.2 Commercial. Hats shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. One hundred and forty-four hats of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A intermediate packs shall be packed three in length, four in width, and one in depth. Inside dimensions of each container shall approximate 44 inches in length, 16-1/2 inches in width, and 14 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. One hundred and forty-four hats of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed three in length, four in width, and one in depth. Inside dimensions of each container shall approximate 44 inches in length, 16-1/2 inches in width, and 14 inches in depth. Approximate dimensions are furnished as a guide only. Each

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shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4.

5.2.3 Commercial packing. Hats, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), hats packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 84 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.3.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The hat is intended for use by military personnel of the Department of Defense for special purposes in desert environments.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

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6.3 Standard samples. For access to sample of the hat and shade samples of the materials, address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

* 6.5 Ventilators with washers. The No. 53 screen type ventilator with washer may be obtained from the Stimpson Company, 900 Sylvan Avenue, Bayport, NY 11705.

* 6.6 Viscosimeter. The Brookfield viscosimeter maybe obtained from Brookfield Engineering Laboratories, Inc., 240 Cushing Street, Stoughten, MA 02072.

* 6.7 Subject term (key word) listing

Camouflage
Clothing, desert
Hat, camouflage

6.8 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA - CT

User activity:

Navy - YD

Preparing activity:

Army - GL

Project No. 8415-0605

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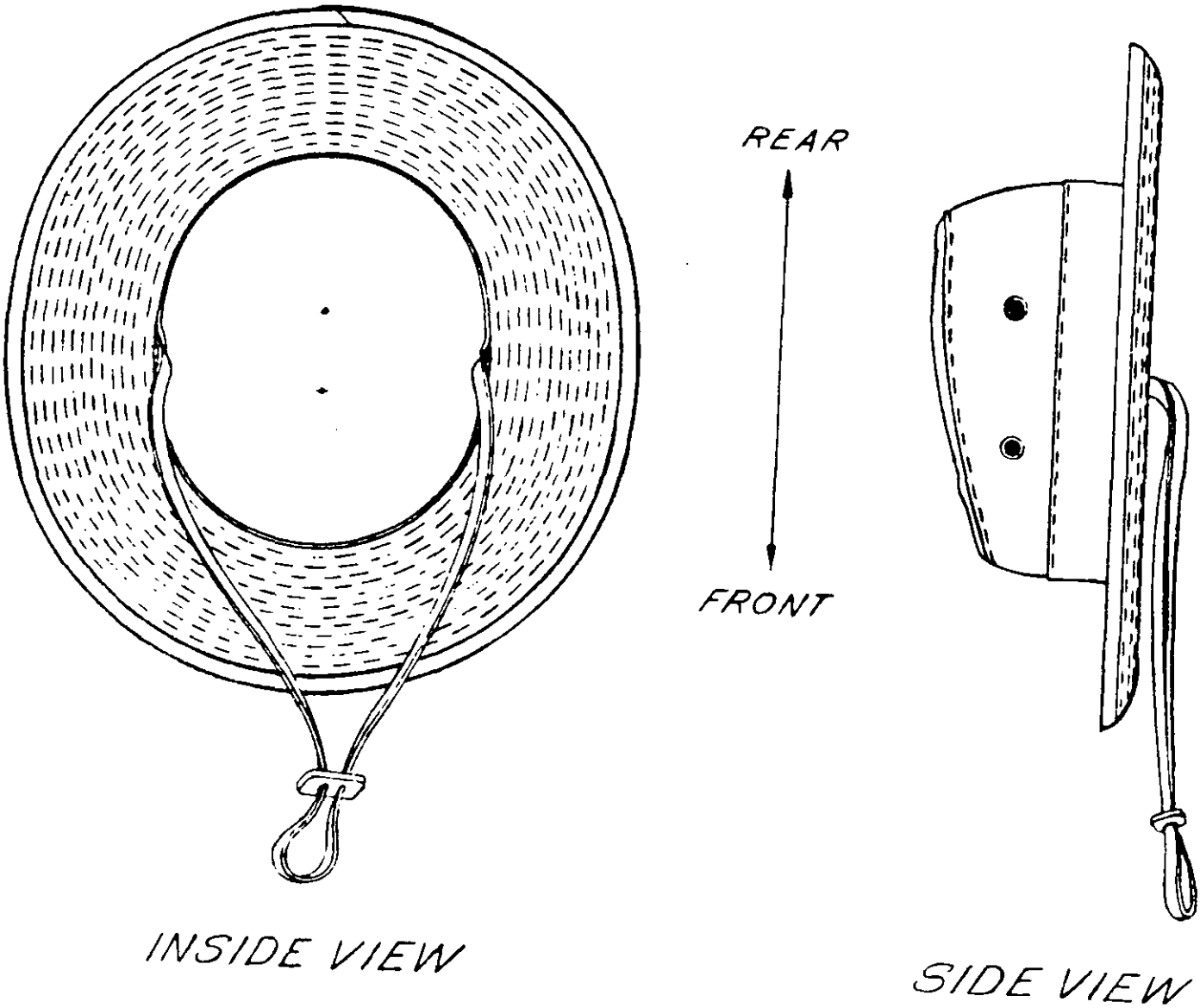


FIGURE 1. Hat, camouflage pattern: desert.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-44105A	Hat, Camouflage Pattern: Desert

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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